APEX AI: End-to-End Predictive Telemetry & Autonomous Strategic Reasoning Architecture

Multi-modal telemetry ingestion, physical aerodynamic clustering, intra-race ranking, and tool-augmented ReAct reasoning

Tier 1: Data Ingestion & Storage

Raw Feeds & Temporal Normalization

64 GPs | 1,119 Finishes

FastF1 & Timing Feeds

- 10 Hz Car Telemetry (GPS, RPM)
- Throttle / Brake / Gear Channels
- Official FIA Timing Intervals

Data Normalizer

`clean_data.py`

- String Intervals to Floats (s)
- Status Flags (finished = 1 / 0)
- Normalized Qualifying Splits

Relational DB Engine

`f1.db (SQLite)`

- race_results (1,278 rows)
- qualifying_results (1,318 rows)
- 2023 - 2025 Multi-Season

Temporal Causality Guard

- Zero Look-Ahead Leakage
- Strictly Ordered Historical Form
- Expanding Train/Test Windows

Filtered SQL Dataset

Tier 2: Feature Engineering

Micro-Sectors & Physics Clustering

23 Domain Features

Micro-Sector Telemetry

- Pace Efficiency: $\eta_{pace} = t_{best}/t_{pole}$
- Progression: $\Pi_{prog} = \Delta t_{Q1 \to Q3}$
- Apex Speed: $\eta_{apex} = V_{min}/(1 + \Delta t)$

Circuit Aerodynamics

`circuit_clusters.py`

- 4 Physics Downforce Clusters
- Full-Throttle Duty: $D_{thr} \in [45, 79]\%$
- Tire Stress: $T_{stress} \in [2.1, 5.0]$
- Overtaking Difficulty: O_{diff}

Driver & Team Baselines

- Rolling Form (Last 3 / All-Time)
- Driver Momentum (Recent vs All)
- Historical Podium / Win Rates
- Teammate Qualifying Delta

Feature Matrix Export

`full_features.csv`

- 1,379 Rows x 40 Columns
- 1,119 Verified Classifications
- Standardized Numerical Tensors

Engineered Features

Tier 3: Supervised ML Core

Predictive Models & Ranking

MAE 2.191 | p = 6.24e-10

Expanding Walk-Forward CV

`walk_forward_cv.py`

- 15 Warmup Events (262 cars)
- 49 Out-of-Sample GP Folds
- 857 Finishes Evaluated
- 19.16% Error Reduction

Intra-Race Ranked XGBoost

`xgb_race_predictor.joblib`

- Continuous Lap Pace Regression
- Discrete Sort: $\hat{P}_i = \text{rank}(\hat{y}_i)$
- MAE: 2.191 (vs 2.711 Baseline)
- Acc ±1 Pos: 49.3%, ±2: 67.2%

High-Stakes Classifiers

- Top-3 Podium Classifier:
  ROC-AUC: 94.43% | Acc: 90.43%
- Top-10 Points Classifier:
  ROC-AUC: 81.78% | Acc: 75.22%

Cluster-Specific Baselines

- Low Downforce Power: 1.942
- High Downforce Thermal: 2.130
- Street Circuit Agility: 2.315

Model Inference & Probs

Re-Query Prediction

Tier 4: ReAct Autonomous Agent

Tool-Augmented Strategic Reasoning

5-Stage Reasoning Loop

ReAct Multi-Step Engine

`react_agent.py`

- Step 1: Baseline Form Profiling
- Step 2: Aerodynamic Regime
- Step 3: Supervised ML Predict
- Step 4: Empirical Reflection
- Step 5: Strategic Thesis Synthesis

Unified Tool Registry

`tools.py`

- 8 Typed JSON Tool Schemas:
  predict_race, circuit_profile,
  driver_career_stats, standings,
  qualifying_race_delta, etc.

Dynamic Sandbox (LATM)

`dynamic_analyst.py`

- LLMs-As-Tool-Makers Pattern
- Safe Python / SQL Execution
- On-the-Fly Telemetry Scripting
- Real-Time Output Interception

Decision Calibration

- Brier Reliability & ECE
- Binomial Test: $z = 9.07, p < 10^{-19}$
- Calibrated Decision Confidence

Domain Guidance & Dynamic SQL

Tier 5: LLM Layer & Strategy

Fine-Tuning & Decision Output

0.52% Parameter QLoRA

Base LLM Model

- Meta Llama-3-8B-Instruct
- 4-Bit NormalFloat (NF4)
- Double Quantization
- 2,048 Sequence Window

QLoRA Fine-Tuning

`f1_technical_alpaca.json`

- LoRA Rank $r = 16, \alpha = 16$
- All 7 Linear Modules
- 41.94M Trainable Params
- Loss: 2.702 -> 0.0031 (-99.9%)
- Fast Unsloth Inference

Live Strategy Synthesis

- Tactical Undercut Timing
- Thermal Cliff Alerts (>130°C)
- Tire Deg Slope Benchmarks
- Monte Carlo Win Probabilities

Publication Brief

- Structured Strategy Memo
- Empirical Audit Trail
- Engineering Tradeoff Matrix